

# SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: DATA ANALYSIS WITH SPSS/SAS/R

## PRACTICAL 7

**AIM:** Selecting and dropping variables using select() in R. import dataset.

### > COMMANDS USED

library(dplyr) # select() is part of the dplyr package

#

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# 1. IMPORT DATASET

#

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```
# Import the CSV file
```

```
ai_jobs <-
```

```
read.csv("AI_Impact_on_Jobs_2030.csv")
```

```
print("--- Original Dataset (First 3 rows) ---")
```

```
print(head(ai_jobs, 3))
```

```
#
```

```
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```

```
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```

```
# 2. SELECTING VARIABLES (Keeping Columns)
```

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#

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**# Method A: Select specific columns by name**

**# Scenario: We only want the job title, salary, and final risk category**

**selected\_cols <- ai\_jobs %>%**

**select(Job\_Title, Average\_Salary, Risk\_Category)**

**print("--- Selected Specific Columns (Job\_Title, Average\_Salary, Risk\_Category) ---")**

**print(head(selected\_cols, 3))**

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**# Method B: Select a range of adjacent columns**

**# Scenario: Select everything from 'Years\_Experience' to 'AI\_Exposure\_Index'**

```
range_cols <- ai_jobs %>%
```

```
select(Years_Experience:AI_Exposure_Index)
```

```
print("--- Selected Range of Columns  
(Years_Experience to  
AI_Exposure_Index) ---")
```

```
print(head(range_cols, 3))
```

**# Method C: Select using helper functions (e.g., starts\_with)**

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```
# Scenario: Select columns that start  
with "S" (Skill_1, Skill_2, etc.)  
starts_with_s <- ai_jobs %>%  
  select(starts_with("S"))
```

```
print("--- Selected columns starting with  
'S' (The 10 Skill columns) ---")  
print(head(starts_with_s, 3))
```

```
#
```

```
-----  
-----
```

```
# 3. DROPPING VARIABLES (Removing  
Columns)
```

```
#
```

```
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```

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**# We use the minus sign (-) to remove variables**

**# Method A: Drop a single specific column**

**# Scenario: Remove the**

**'Tech\_Growth\_Factor'**

**dropped\_one <- ai\_jobs %>%**

**select(-Tech\_Growth\_Factor)**

**print("--- Dataset with**

**'Tech\_Growth\_Factor' dropped (Names to verify) ---")**

**print(names(dropped\_one))**

**# Method B: Drop multiple columns**

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**# Scenario: Remove 'Years\_Experience' and 'Education\_Level'**

```
dropped_multiple <- ai_jobs %>%  
  select(-Years_Experience,  
-Education_Level)
```

```
print("--- Dataset with  
'Years_Experience' and  
'Education_Level' dropped (Names to  
verify) ---")
```

```
print(names(dropped_multiple))
```

**# Method C: Drop a range of columns**

**# Scenario: Remove all 10 Skill columns  
(from 'Skill\_1' to 'Skill\_10')**

```
dropped_range <- ai_jobs %>%  
  select(-(Skill_1:Skill_10))
```

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```
print("--- Dataset with range 'Skill_1' to  
'Skill_10' dropped (Names to verify) ---")  
print(names(dropped_range))
```

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```
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AI Impact on Jobs 2030
Job Title Average Salary Years Experience Education Level AI Exposure Index Tech Growth Factor Automation Probability 2030 Risk Category

> library(dplyr) # select() is part of the dplyr package
> # 1. IMPORT DATASET
> # Import the CSV file
> ai_jobs <- read.csv("AI_Impact_on_Jobs_2030.csv")
> print("---- Original Dataset (First 3 rows) ----")
> print(head(ai_jobs, 3))
  Job_Title Average_Salary Years_Experience Education_Level AI_Exposure_Index Tech_Growth_Factor Automation_Probability_2030 Risk_Category
1 Security_Guard 45795 28 Master's 0.18 1.28
2 Research_Scientist 133355 20 PhD 0.62 1.11
3 Construction_Worker 146216 2 High_School 0.86 1.18
  Skill_1 Skill_2 Skill_3 Skill_4 Skill_5 Skill_6 Skill_7 Skill_8
1 0.45 0.10 0.46 0.33 0.14 0.65 0.06 0.72
2 0.02 0.52 0.40 0.05 0.97 0.23 0.09 0.62
3 0.02 0.94 0.56 0.39 0.02 0.23 0.24 0.68
  Skill_9 Skill_10
1 0.94 0.00
2 0.38 0.98
3 0.61 0.83
> # 2. SELECTING VARIABLES (Keeping Columns)
> # Method A: Select specific columns by name
> # Scenario: We only want the job title, salary, and final risk category
> selected_cols <- ai_jobs %>%
+ select(Job_Title, Average_Salary, Risk_Category)
> print("---- Selected Specific Columns (Job_Title, Average_Salary, Risk_Category) ----")
> print(head(selected_cols, 3))
  Job_Title Average_Salary Risk_Category
1 Security_Guard 45795 High
2 Research_Scientist 133355 Low
3 Construction_Worker 146216 High
> # Method B: Select a range of adjacent columns
> # Scenario: Select everything from 'Years_Experience' to 'AI_Exposure_Index'
> range_cols <- ai_jobs %>%
+ select(Years_Experience:AI_Exposure_Index)
> print("---- Selected Range of Columns (Years_Experience to AI_Exposure_Index) ----")
> print(head(range_cols, 3))
  Years_Experience Education_Level AI_Exposure_Index
1 28 Master's 0.18
2 20 PhD 0.62
3 2 High_School 0.86
> # Method C: Select using helper functions (e.g., starts_with())
> # Scenario: Select columns that start with 'S' (Skill_1, Skill_2, etc.)
> starts_with_s <- ai_jobs %>%
+ select(starts_with("S"))
> print("---- Selected columns starting with 'S' (The 10 Skill columns) ----")
> print(head(starts_with_s, 3))
  Skill_1 Skill_2 Skill_3 Skill_4 Skill_5 Skill_6 Skill_7 Skill_8 Skill_9 Skill_10
1 0.45 0.10 0.46 0.33 0.14 0.65 0.06 0.72 0.94 0.00
2 0.02 0.52 0.40 0.05 0.97 0.23 0.09 0.62 0.38 0.98
3 0.02 0.94 0.56 0.39 0.02 0.23 0.24 0.68 0.61 0.83
> # 3. DROPPING VARIABLES (Removing Columns)
> # We use the minus sign (-) to remove variables
> # Method A: Drop a single specific column
> # Scenario: Remove the 'Tech_Growth_Factor'
> dropped_one <- ai_jobs %>%
+ select(-Tech_Growth_Factor)
> print("---- Dataset with 'Tech_Growth_Factor' dropped (Names to verify) ----")
> print(names(dropped_one))
[1] "Job_Title" "Average_Salary" "Years_Experience"
[4] "Education_Level" "AI_Exposure_Index" "Automation_Probability_2030"
[7] "Risk_Category" "Skill_1" "Skill_2"
[10] "Skill_3" "Skill_4" "Skill_5"
[13] "Skill_6" "Skill_7" "Skill_8" "Skill_9" "Skill_10"
```

```
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AI Impact on Jobs 2030
Job Title Average Salary Years Experience Education Level AI Exposure Index Tech Growth Factor Automation Probability 2030 Risk Category

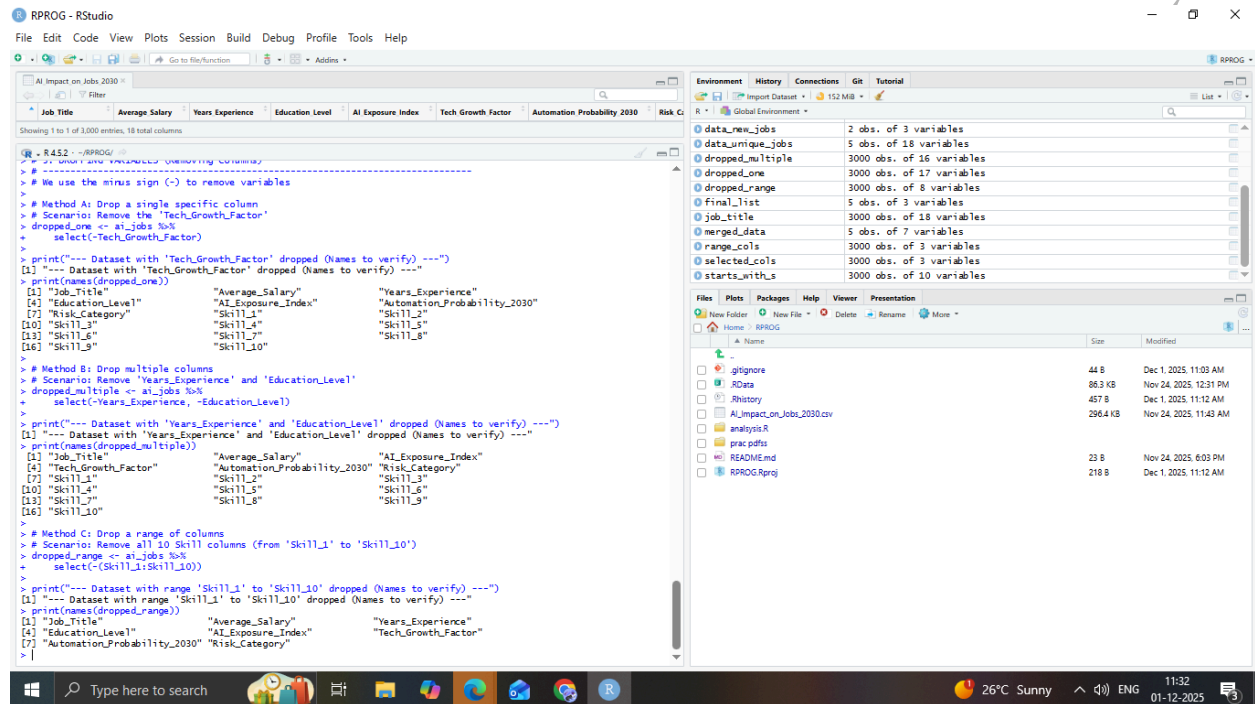
> # Method B: Select a range of adjacent columns
> # Scenario: Select everything from 'Years_Experience' to 'AI_Exposure_Index'
> range_cols <- ai_jobs %>%
+ select(Years_Experience:AI_Exposure_Index)
> print("---- Selected Range of Columns (Years_Experience to AI_Exposure_Index) ----")
> print(head(range_cols, 3))
  Years_Experience Education_Level AI_Exposure_Index
1 28 Master's 0.18
2 20 PhD 0.62
3 2 High_School 0.86
> # Method C: Select using helper functions (e.g., starts_with())
> # Scenario: Select columns that start with 'S' (Skill_1, Skill_2, etc.)
> starts_with_s <- ai_jobs %>%
+ select(starts_with("S"))
> print("---- Selected columns starting with 'S' (The 10 Skill columns) ----")
> print(head(starts_with_s, 3))
  Skill_1 Skill_2 Skill_3 Skill_4 Skill_5 Skill_6 Skill_7 Skill_8 Skill_9 Skill_10
1 0.45 0.10 0.46 0.33 0.14 0.65 0.06 0.72 0.94 0.00
2 0.02 0.52 0.40 0.05 0.97 0.23 0.09 0.62 0.38 0.98
3 0.02 0.94 0.56 0.39 0.02 0.23 0.24 0.68 0.61 0.83
> # 3. DROPPING VARIABLES (Removing Columns)
> # We use the minus sign (-) to remove variables
> # Method A: Drop a single specific column
> # Scenario: Remove the 'Tech_Growth_Factor'
> dropped_one <- ai_jobs %>%
+ select(-Tech_Growth_Factor)
> print("---- Dataset with 'Tech_Growth_Factor' dropped (Names to verify) ----")
> print(names(dropped_one))
[1] "Job_Title" "Average_Salary" "Years_Experience"
[4] "Education_Level" "AI_Exposure_Index" "Automation_Probability_2030"
[7] "Risk_Category" "Skill_1" "Skill_2"
[10] "Skill_3" "Skill_4" "Skill_5"
[13] "Skill_6" "Skill_7" "Skill_8" "Skill_9" "Skill_10"
```

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The screenshot displays the RStudio interface. The main editor shows R code for data manipulation. The Environment pane on the right lists the objects created during the session.

```
R - R432 - RPROG
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AI Impact on Jobs 2030
Showing 1 to 1 of 1,000 entries, 18 total columns
Job Title Average Salary Years Experience Education Level AI Exposure Index Tech Growth Factor Automation Probability 2030 Risk Category

> # Method A: Drop a single specific column
> # Scenario: Remove the 'Tech_Growth_Factor'
> dropped_one <- ai_jobs %>%
+   select(-Tech_Growth_Factor)
>
> print("---- Dataset with 'Tech_Growth_Factor' dropped (Names to verify) ----")
[1] "---- Dataset with 'Tech_Growth_Factor' dropped (Names to verify) ----"
> print(names(dropped_one))
[1] "Job_Title" "Average_Salary" "Years_Experience"
[4] "Education_Level" "AI_Exposure_Index" "Automation_Probability_2030"
[7] "Risk_Category" "Skill_1" "Skill_2"
[10] "Skill_3" "Skill_4" "Skill_5"
[13] "Skill_6" "Skill_7" "Skill_8"
[16] "Skill_9" "Skill_10"
>
> # Method B: Drop multiple columns
> # Scenario: Remove 'Years_Experience' and 'Education_Level'
> dropped_multiple <- ai_jobs %>%
+   select(-Years_Experience, -Education_Level)
>
> print("---- Dataset with 'Years_Experience' and 'Education_Level' dropped (Names to verify) ----")
[1] "---- Dataset with 'Years_Experience' and 'Education_Level' dropped (Names to verify) ----"
> print(names(dropped_multiple))
[1] "Job_Title" "Average_Salary" "AI_Exposure_Index"
[4] "Tech_Growth_Factor" "Automation_Probability_2030" "Risk_Category"
[7] "Skill_1" "Skill_2" "Skill_3"
[10] "Skill_4" "Skill_5" "Skill_6"
[13] "Skill_7" "Skill_8" "Skill_9"
[16] "Skill_10"
>
> # Method C: Drop a range of columns
> # Scenario: Remove all 10 Skill columns (from 'Skill_1' to 'Skill_10')
> dropped_range <- ai_jobs %>%
+   select(-(Skill_1:Skill_10))
>
> print("---- Dataset with range 'Skill_1' to 'Skill_10' dropped (Names to verify) ----")
[1] "---- Dataset with range 'Skill_1' to 'Skill_10' dropped (Names to verify) ----"
> print(names(dropped_range))
[1] "Job_Title" "Average_Salary" "Years_Experience"
[4] "Education_Level" "AI_Exposure_Index" "Tech_Growth_Factor"
[7] "Automation_Probability_2030" "Risk_Category"
> |
```

**Environment**

| Object           | Size                      |
|------------------|---------------------------|
| data_new_jobs    | 2 obs. of 3 variables     |
| data_unique_jobs | 5 obs. of 18 variables    |
| dropped_multiple | 3000 obs. of 16 variables |
| dropped_one      | 3000 obs. of 17 variables |
| dropped_range    | 3000 obs. of 8 variables  |
| Final_list       | 5 obs. of 3 variables     |
| job_title        | 3000 obs. of 18 variables |
| merged_data      | 5 obs. of 7 variables     |
| range_cols       | 3000 obs. of 3 variables  |
| selected_cols    | 3000 obs. of 3 variables  |
| starts_with_s    | 3000 obs. of 10 variables |

**Files**

| Name                       | Size     | Modified               |
|----------------------------|----------|------------------------|
| gignore                    | 44 B     | Dec 1, 2025, 11:03 AM  |
| RData                      | 86.3 KB  | Nov 24, 2025, 12:31 PM |
| Rhistory                   | 457 B    | Dec 1, 2025, 11:12 AM  |
| AI_Impact_on_Jobs_2030.csv | 296.4 KB | Nov 24, 2025, 11:43 AM |
| analysis.R                 |          |                        |
| prac.pdfs                  |          |                        |
| README.md                  | 23 B     | Nov 24, 2025, 6:03 PM  |
| RPROG.Rproj                | 218 B    | Dec 1, 2025, 11:12 AM  |

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